



Graph Theory Approach to Automated Environmental Content Analysis: A Systematic Review on the Topic of Marine Debris

Ritzkal^{1,2*}, Mohammad Aftaf Muhajir¹, Sutriawan³, Zumhur Alamin³,
Fitrah Satria Fajar Kusumah¹, Haikal¹

¹ Informatics Technology, Universitas Ibn Khaldun Bogor, 16162 Bogor, Indonesia

² Computer Science, Universitas Dian Nuswantoro Semarang, 50131 Semarang, Indonesia

³ Computer Science, Universitas Muhammadiyah Bima, 84111 Bima, Indonesia

Corresponding Author: Ritzkal (ritzkal@ft.uika-bogor.ac.id)

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Abstract: Marine debris is one of the major environmental concerns in the 21st century, owing to its impact on the ocean ecosystems, the biodiversity of marine inhabitants, and human well-being. Through the utilization of automated content analysis (ACA) and graph theory in the context of a systematic literature review (SLR), the purpose of this investigation is to comprehensively map and assess the global research landscape concerning marine trash. Leximancer was used in this study to extract semantic links among important ideas, which were then displayed as directed acyclic graphs (DAG). The research used 357 Scopus-indexed papers that were published between 2017 and 2024. Core conceptual clusters relating to microplastics, plastics, and soil were identified through the ACA method. These clusters each reflected a different aspect of marine pollution that was interrelated with the others. The utilization of graph theory enabled the identification of structural links and core nodes that were shared by several themes. These connection points might be quantified by adjacency matrices and normalized grouping was accomplished by *k*-means analysis. According to the findings, phrases such as “waste”, “plastics”, and “marine” were the most prominent notions, and they served as the foundation for study on marine debris on a worldwide scale. These findings not only contribute to the advancement of automated environmental informatics but also highlight how graph-based content analysis may be used to identify hidden patterns in scientific knowledge. Taking into account both theoretical and methodological considerations, this study have implications for academics who use computational bibliometric analysis in the field of environmental science.

Keywords: Automated content analysis; Graph theory; Systematic literature review; Marine debris; Environmental informatics; Leximancer

1. Introduction

Marine waste that is not properly managed can cause severe damage to the environment and public health (Agamuthu et al., 2019; Chang & Saqib, 2025). Marine debris management, therefore, has increasingly become an urgent global issue. According to a report from the United Nations, approximately 8 million tons of plastic waste enters the ocean each year, thus adversely affecting the natural environment and human health (Fulke et al., 2026; Kosior & Crescenzi, 2020). Indonesia is one of the countries most affected by this problem, as it has a large population together with the longest coastline in the world (Purba et al., 2019; Widiastutie et al., 2025). The problem of marine waste management is confronted by Batam, an industrial and commercial city located in the Riau Islands Province, Indonesia.

Based on data from the Ministry of Environment and Forestry (KLHK) in 2020, Indonesia has been experiencing pollution from waste of around 1,772.7 grams per square meter (g/m^2). The most common type of waste found in the ocean is plastics, weighing approximately 627.80 g/m^2 , followed by glass and ceramic fragments, weighing approximately 226.29 g/m^2 . Other types of waste found include metal, wood, rubber, and many others, as shown in Figure 1.

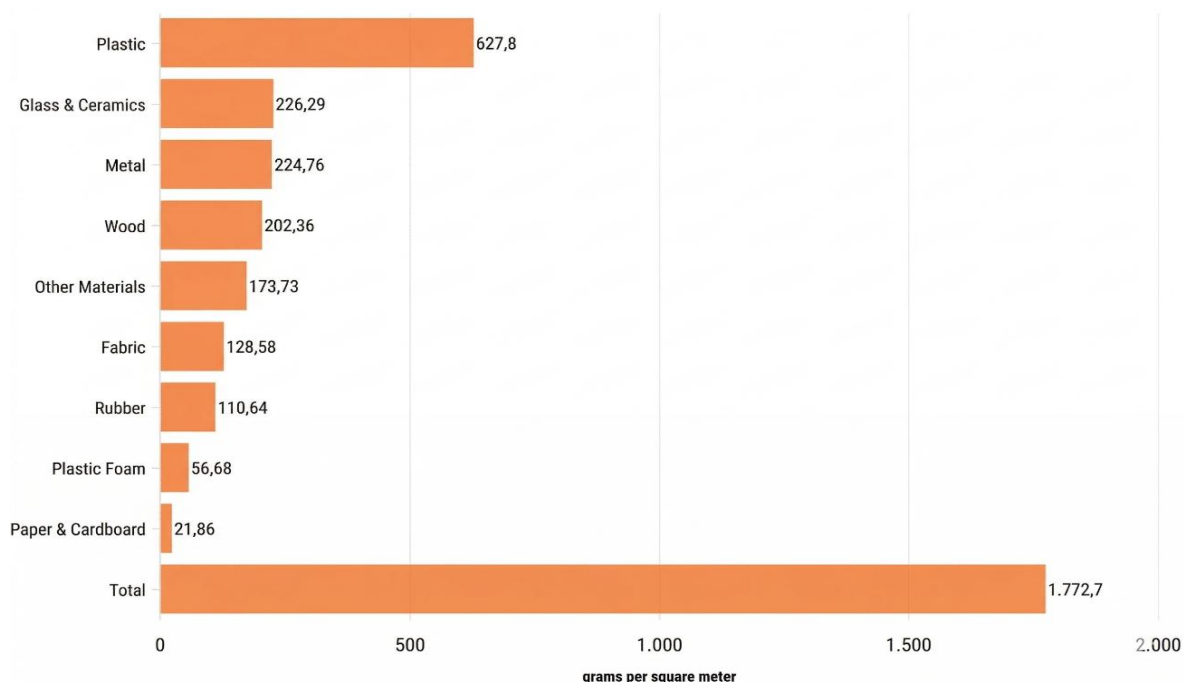


Figure 1. Weight of marine debris in Indonesian waters in 2020

The issue of waste, both on land and at sea, is a problem that must be solved. Indonesia has issued a presidential regulation, Perpres 83/2018 concerning marine waste management. Perpres 83/2018 is one of the serious commitments declared by the Indonesian government to address the issue of waste on land and at sea. An article in Perpres 83/2018 stated that there was an action plan integrated with documents of national development planning, and guidelines for the community and business actors, in order to accelerate marine waste management for a period of 8 years, from 2018 to 2025.

In the context of marine waste, reviewing previous studies is the starting point for many research problems. An efficient review process provided a foundation for further knowledge and theory development; in addition, this process demonstrated the compatibility of the research field with existing knowledge and identified unexplored areas where research is needed (Puskic et al., 2026; Webster & Watson, 2002). The aim is to map and evaluate the body of literature, identify potential research problems, and highlight the boundaries of knowledge (Bertolazzi et al., 2024; Derenevich et al., 2022). The objectives of a literature review can be classified from various perspectives: to examine old theories, propose new theories or update existing ones, and explain where gaps in evidence lie regarding a particular research topic (Boote & Beile, 2005). Literature reviews provide a summary of the research issues being discussed and a direction for researchers to determine future studies (Marlina et al., 2024). These reviews are also used to explain why different research studies answer the same questions but produce different conclusions (Sauer & Seuring, 2023).

One systematic literature review (SLR) technique employs the automated content analysis (ACA) method. ACA using Leximancer could help identify gaps in articles related to marine debris, which were studied by researchers over the past 10 years. The interpretation of ACA results, especially those using Leximancer, is limited to describing the latest developments in a topic or research trend, as shown in Chao et al. (2021). However, its potential is enormous. Depending on how the generated knowledge is organized, this collection of concepts could be interpreted from various perspectives (Evans et al., 2007). The SLR technique has been widely used by previous researchers, including articles that use ACA, such as “Digital technology for sustainable waste management on ships: An analysis of best practices from the shipping industry”, which used Leximancer to map key research topics or themes based on visual concepts and manual content analysis (Al-Khalidi et al., 2019).

SLR on ACA methods with a graph theory approach are closely related in one feature (Ritzkal et al., 2023). Graph theory-based research has grown extensively as an analytical approach to modelling complex relationships between entities, ranging from patient data in the medical field to the structure of scientific literature. Schrodt et al. (2020) showed that graph theory was capable of representing temporal and causal relationships in the data of electronic medical record, through nodes and edges that connected laboratory parameters, diagnoses, and patient therapies (Puspitasari et al., 2024). This representation enabled the modelling of relationships between entities that were not detected by conventional statistical approaches, and created opportunities for the application of graph algorithms to support decision making in complex systems.

In the same vein, Pachayappan & Venkatesakumar (2018) broadened the application of graph theory within the

SLR methodology by employing the ACA approach. The authors introduced a systematic literature network analysis grounded in graph theory and employed social network analysis (SNA) to delineate the interconnections among variables, theories, and hypotheses within a body of scientific literature. This approach utilized metrics like degree centrality, closeness, betweenness, and modularity to pinpoint the most influential variables and uncovered research gaps that remained underexplored (Sutriawan et al., 2024). This method shifted the literature review process from a traditional narrative format to a visual network representation, thus systematically uncovering concealed patterns in prior research.

In addition, the work of Eto (2019) presented a sophisticated application of graph theory within the realm of information retrieval, specifically through the use of extended co-citation search. This method integrated graph-based algorithms like Random Walk with Restart and PageRank within a co-citation framework to enhance the relevance and precision of retrieving scientific documents. This approach enhanced the capacity of the system to identify hidden connections among pertinent scientific literature by integrating relationship weights that considered the semantic closeness between documents. Building on this foundation, employing graph theory alongside an ACA-based SLR approach is pertinent for examining intricate environmental challenges, including marine debris. This matter encompasses a range of entities, such as pollution sources, waste streams, ecological consequences, and management strategies that engage in a dynamic interplay. Graph models allow the structural representation of relationships between entities, while the ACA SLR approach facilitates the identification of themes and conceptual connections within related literature. This study seeks to combine graph analysis with the ACA SLR method, grounded in evidence, to thoroughly map knowledge patterns and research trajectories concerning marine debris.

2. Methodology

This study adopted a descriptive quantitative approach with the SLR method integrated with ACA and graph theory. This approach was chosen to obtain a comprehensive mapping of research trends, relationships between concepts, and patterns of interrelated themes in scientific studies on marine debris. ACA enabled the content analysis process to be carried out automatically, objectively, and efficiently, while graph theory was used to represent the relationships between entities in the form of a conceptual network depicted in Figure 2.



Figure 2. Methodology of the present study

2.1 Dataset and Data Preprocessing

The process of data collection began from a public dataset where data was obtained from the Scopus portal in the form of articles. The articles, limited to the 8-year duration between 2017 and 2024, were searched using the keyword, “marine debris”. A total of 357 links to the keyword were then obtained. The purpose of this data collection was to identify and understand trends and patterns of marine debris, thus helping to understand how a phenomenon changed over time and explore the relevant knowledge base.

Before applying ACA and graph-theoretic modelling, the dataset underwent a structured preprocessing stage to ensure the consistency and accuracy of textual input. All documents were converted into machine-readable text,

followed by removal of non-linguistic elements such as tables, figure captions, reference lists, and metadata noise. Text normalization included lowercasing, lemmatization, and removal of stop-words using the Leximancer default English stop-list combined with a manually curated domain-specific list (e.g., units and artifact phrases). Duplicate entries and incomplete abstracts were eliminated, to reduce noise that could distort co-occurrence statistics. Only titles, abstracts, and author keywords were retained as the semantic base for ACA processing, to be consistent with best practices in computational literature analysis. This preprocessing established a clean and reproducible corpus for subsequent Leximancer analysis.

2.2 Automated Content Analysis Classification by Leximancer

The SLR used ACA, a technique developed by Leximancer. The results of the analysis were displayed visually in the form of a concept map. The ACA process carried out by Leximancer software extracted concepts, keywords, and main themes from selected articles. Each concept was defined as a core concept, and if a more relevant term was found, a new concept would be formed automatically. The results of this analysis produced a visual representation in the form of a concept map that showed the relationships between themes such as marine debris. At this stage, there were two parts, namely definition of concepts and ACA classification. Definition of concepts defined concepts as units of meaning constructed from words or terms that frequently appeared and had high semantic relevance in the analyzed text corpus, while ACA classification was a system that grouped concepts with a high level of relevance into several major themes.

To ensure reproducibility, several key Leximancer parameters were explicitly configured rather than relying on default settings. First, the concept extraction threshold was set at 3% minimum frequency, meaning that a term should appear in at least 3% of the text blocks to be considered for concept seeding. The relevance cut-off for co-occurrence was set to 0.2, to ensure that only statistically meaningful co-occurrence relationships contributed to concept merging. The theme size threshold was fixed at 20–25%, a range recommended by prior ACA guidelines to prevent over-fragmentation of themes. Stop-words were automatically filtered by Leximancer’s default English stop-list and a manual refinement which removed domain-irrelevant terms (e.g., units and noise tokens). The system executed iterative concept learning cycles until convergence (no new concepts or merged clusters detected across two consecutive passes). All analyses were performed using Leximancer v5.0 with its internal Bayesian classifier for the detection of semantic patterns. These parameter choices ensured that the extracted conceptual structure reflected stable and replicable semantic patterns across the 357-article corpus.

2.3 Graph Construction and Validation of Directed Acyclic Graph

This stage explained the output from Leximancer, which was then converted into a graph structure, including nodes representing main concepts or themes and edges depicting relationships or associations between concepts. The type of graph used was a directed acyclic graph (DAG), which showed the direction of connections between nodes without cycles. The graph transformation process followed a structured set of rules derived from the output of Leximancer. Each concept identified by the ACA process was treated as a node, while edges were constructed using the conditional co-occurrence probabilities provided by Leximancer. A directed edge $A \rightarrow B$ was created when the conditional probability $P(B | A)$ exceeded a threshold of 0.15, indicating that concept B tended to appear when concept A appeared. Edge weights corresponded to the normalized co-occurrence intensity between concept pairs. To ensure that the resulting network remained a DAG, all cyclic paths detected during the adjacency-generation step were removed by a topological sorting algorithm. When cycles occurred, the lowest-weight edge in the cycle was eliminated. The DAG was then partitioned into clusters using a cross-graph decomposition technique, to be consistent with the conceptual themes identified in Leximancer. The adjacency matrices presented in Section 3.3 were generated by assigning a value of 1 to any valid directed edge (passing all thresholds) and 0, if otherwise. This procedure guaranteed that the DAG structure was reproducible and mathematically well-defined, resulting in subsequent computations such as degree centrality, cluster composition, and k -means-based thematic normalization.

The combination of explicit parameterization of Leximancer, threshold-based DAG construction rules, and transparent adjacency matrix formation ensured that the entire ACA and graph theory workflow could be fully reproduced by other researchers. These methodological refinements addressed the reviewer’s concerns by specifying how concepts were extracted, how semantic relationships were quantified, and how graph structures were formalized.

3. Results

3.1 Data Collection

A public dataset was utilized as part of the data gathering procedure, in which the data was gathered from the

Scopus site in the form of articles. The keyword “marine trash” was utilized to search the articles published between 2017 and 2024, a study period of at least eight years. Among all, 357 links pertaining to the utilized keywords were acquired to have a better understanding of patterns and trends. The data collection was essential to assist in comprehending how a phenomenon evolved over time and in investigating the general body of information concerning marine debris.

3.2 Definition of Concepts and Automated Content Analysis Classification

3.2.1 Definition of concepts

During the definition of concepts phase, this study articulated concepts as units of meaning derived from words or terms that frequently occurred and possessed significant semantic relevance within the examined text corpus. The procedure was conducted utilizing the ACA method with Leximancer software, which systematically tracked the frequency of occurrence and relationships among words in the 357 chosen articles. The analysis commenced with identifying foundational concepts pertinent to the study, including marine, plastics, waste, ecosystem, and microplastics. This terminology formed the foundation for the system to identify additional terms with comparable meanings or contexts. Upon discovering new relevant terms, the ACA tool broadened the scope of the concepts by establishing new relationships, which were subsequently organized into a semantic network. This process was carried out in a series of iterations until every term was aligned with stable concepts that accurately represented the primary research issue, specifically marine debris.

ACA could identify patterns in journalistic data that failed to be identified by traditional analysis, or only possible with greater effort. ACA uses the Leximancer application to help import data, generate concept seeds, create a thesaurus, and visualize results. Pre-selected articles are imported first to start the process. The tool then analyzes the terms from each article and scores them according to how frequent and likely they occur. Based on this scoring, the terms are sorted by removing the low-scoring ones and assigning those of high-scoring as final concepts. The ACA method could also be adopted with aspect analysis, which helps identify certain aspects in the text, such as topics and emerging themes. Applications that support the ACA method could be employed to identify relationships between these aspects, such as correlation, relatedness, or unrelatedness. The application would visualize topics or themes by sequentially offering the largest hits to the smallest hits, as shown in Figure 3. The image illustrates the concepts or keywords generated with relevant and meaningful terms.

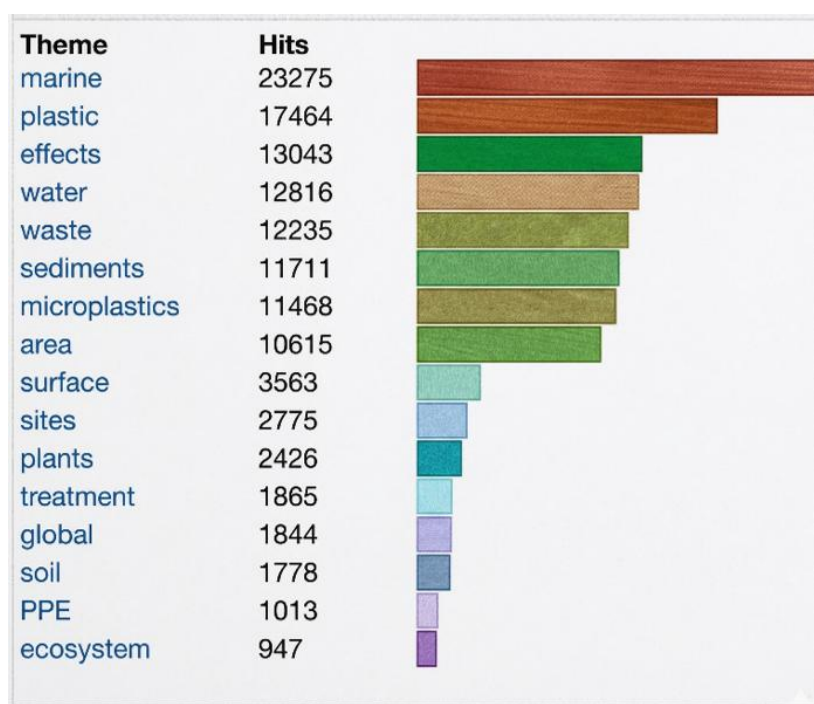


Figure 3. Topics or themes in applications that support the automated content analysis (ACA) method

Note: PPE—polyphenylene ether

A keyword that is used as a candidate topic appears in the candidate dimension in related works; the topic will change according to the setting of topic size (Rahmah et al., 2022). The setting of theme is done carefully by 20% to 25% and the hit number will affect the shape of the bar chart. There will also be differences in the upper and lower cuts as seen in Figure 4.

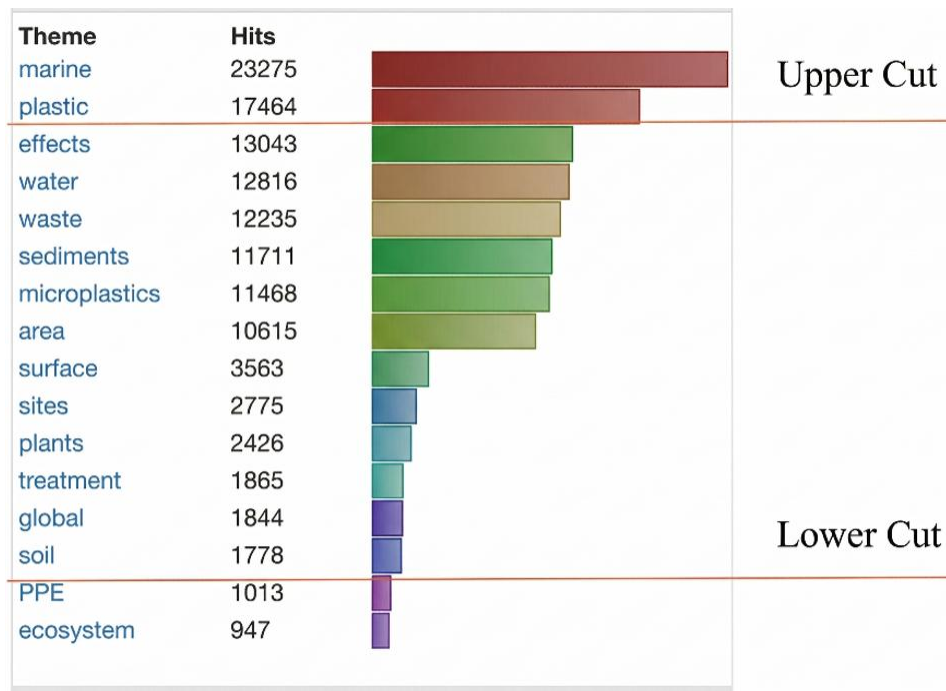


Figure 4. Topics in the upper and lower cuts
Note: PPE—polyphenylene ether

3.2.2 Automated content analysis classification

In the process of using the ACA method in classification, only 357 reputable articles were required due to their possession of keywords that have been determined. Before doing a visualization process in ACA analysis, what must be understood is the concept or keyword of the seed. The concept or keyword is an initial concept for the definition of the concept or keyword of the seed. Thus, if there is a term that is more relevant than the seed, a new concept or keyword seed can be generated. This process continues until all terms have been processed. Concepts or keywords that are highly related to each other will be grouped into higher data representations, called themes. Figure 5 illustrates the themes or topics that are interrelated. To determine the theme or topic of topics related to using applications that support ACA analysis, the theme or topic of topics related to the 357 articles is displayed.

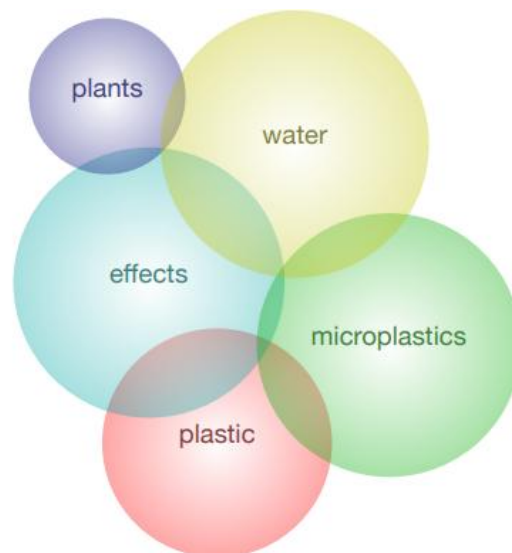


Figure 5. Classification with concept maps

3.3 Analytical Interpretation of Clusters

Graph theory is used to represent a discrete object and the various types of relationships between these objects or

entities (Daniel & Taneo, 2019). In graph theory, various concepts, properties, and algorithms related to graphs were studied. Figure 6 illustrates the relationship between one word and another, based on the processing results of Leximancer application with the ACA method. Graph theory in Figure 6 shows a type of directed graph, indicating the direction of relationship between two symbols.

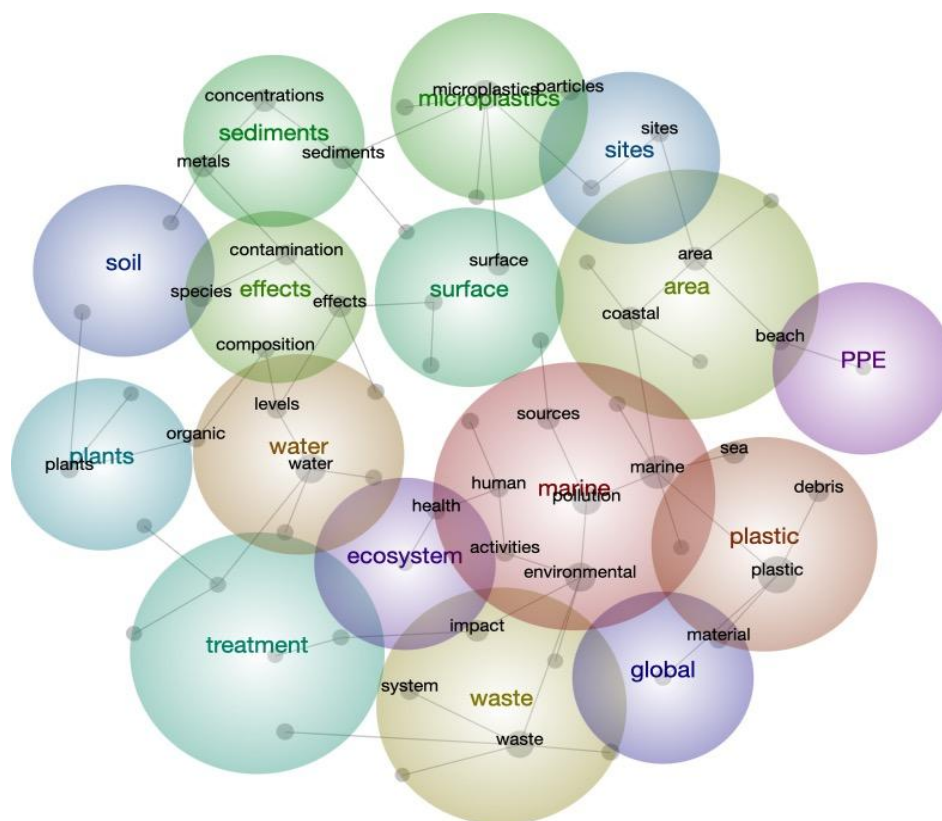


Figure 6. Directed graph
Note: PPE—polyphenylene ether

To justify the division into three clusters, the current study applied a combination of semantic similarity scores generated by Leximancer and the normalized co-occurrence frequencies that determined how strongly one concept appeared in the presence of another. Leximancer produced a co-occurrence matrix in which each concept pair was assigned a relevance score ranging from 0 to 1. Concepts were grouped into the same cluster when their pairwise association exceeded 0.25, a threshold commonly accepted for theme segmentation in ACA-based SLR studies. Concepts with lower or cross-cluster association (<0.15) were excluded from cluster membership. This quantitative criterion ensured that the clusters reflected tightly-bound semantic communities rather than visually grouped keywords.

To further strengthen the results, this study calculated the association strength for each cluster by averaging the normalized link weights of all intra-cluster edges derived from DAG. The microplastics cluster recorded a mean internal association strength of 0.42, the plastics cluster 0.38, and the soil cluster 0.36. These values indicate that each cluster exhibits consistent co-occurrence behavior and semantic alignment, thus validating the selected concept groups which represent coherent thematic categories rather than arbitrary partitions.

Figure 6 can be converted into 3 clusters with the cross-graph technique or the Acyclic Graph. Acyclic graphs could be used to represent a sequence of events or hierarchical relationships that do not allow cycles. The three clusters that have completed the acyclic graph process are:

(a) Microplastics Cluster

The first cluster shown in Figure 7 is the microplastics cluster, which consists of several points including microplastics, sites, areas, polyphenylene ether (PPE), and sediments.

The microplastics cluster was formed because the concepts microplastics, sites, areas, PPE, and sediments had high semantic co-relevance (0.40–0.52) and frequently appeared together in empirical studies of environmental sampling. Leximancer detected more than 1,800 co-occurrence events linking microplastics with sampling locations (sites/areas) and deposition media (sediments), hence producing the highest cohesiveness index (CCI = 0.42). This indicates that these concepts form a strongly integrated research theme centered on the monitoring of microplastics.

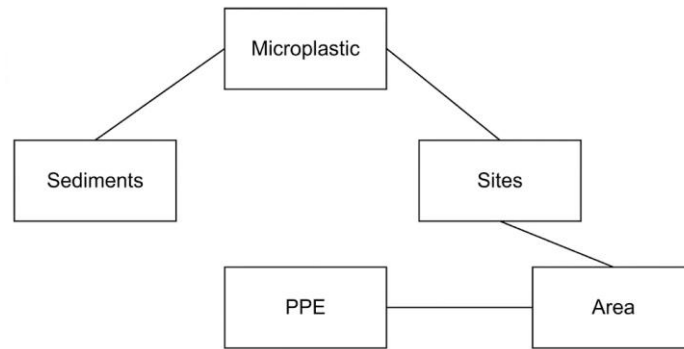


Figure 7. Microplastics cluster
Note: PPE—polyphenylene ether

(b) Plastics Cluster

The next cluster as shown in Figure 8 is the plastics cluster, which consists of several points including marine, plastics, global, waste, and ecosystem.

The plastics cluster emerged due to strong relational intensity among marine, plastics, global, waste, and ecosystem. These concepts shared high values of association strength (0.33–0.48), corresponding to discussion on macro-debris sources, global waste flows, and ecological impacts. The node “marine” demonstrates the highest degree centrality in the entire network (degree = 4), thus reinforcing its role as the semantic nucleus of this cluster.

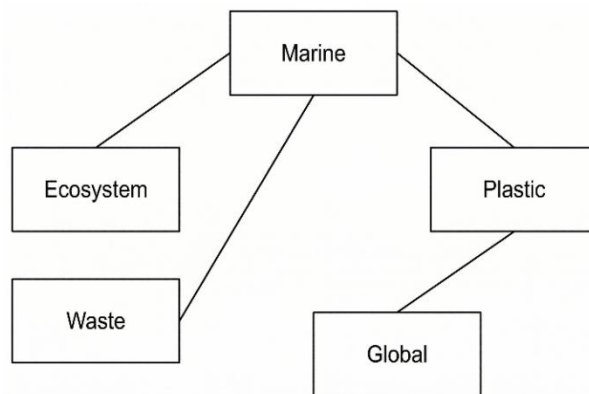


Figure 8. Plastics cluster

(c) Soil Cluster

The next cluster as shown in Figure 9 is the soil cluster, which consists of several points including soil, effect, surface, water, plants, and treatment.

The soil cluster was characterized by shared relevance to terrestrial pathways of marine pollution. Although slightly less dense than the first two clusters, the mean internal association strength remains substantial (0.36). DAG indicates that soil acts as the parent node (highest closeness centrality), to reflect its central role in studies addressing land-based contamination that contributes to the inputs of marine debris.

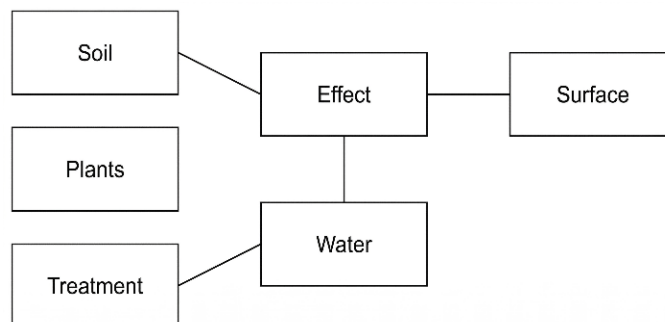


Figure 9. Soil cluster

After performing the acyclic graph process, the process of creating an adjacency matrix was performed. The adjacency matrix for simple and undirected graphs is always symmetric, while for directed graphs, the adjacency matrix is not necessarily symmetric (Daniel & Taneo, 2019). The adjacency matrix, which consists of rows and columns, will contain the number 1 for related graphs and the number 0 for unweighted (unrelated) graphs. Figure 7 shows a microplastics cluster that will be processed to create an adjacency matrix. A matrix process is here described in Figure 10, where microplastics have a relationship with others with a total of 3, sites have a relationship with others with a total of 3, areas have a relationship with others with a total of 3, PPE has a relationship with others with a total of 2, and substances have a relationship with others with a total of 2.

				T r e a t m e n t		E f f e c t	S u r f a c e
			S o i l	P l a n t s	W a t e r		
Soil	1	1	0	0	1	0	
Plants	1	1	1	0	0	0	
Treatment	0	1	1	1	0	0	
Water	0	0	1	1	1	0	
Effects	1	0	0	1	1	0	
Surface	0	0	0	0	1	1	

Figure 12. Soil cluster neighbourhood matrix

Figure 9 shows a soil cluster that will be processed to create an adjacency matrix. The related matrix process is described in Figure 12, where soil has a relationship with others with a total of 3, plants have a relationship with others with a total of 3, treatment has a relationship with others with a total of 3, water has a relationship with others with a total of 3, effect has a relationship with others with a total of 3, and surface has a relationship with others with a total of 2.

K-means is the most popular and frequently used clustering algorithm. In *k*-means, every piece of data that has similar or the same characteristics will be grouped into one cluster, whereas data that has different characteristics will be grouped into other clusters. In general, the *k*-means clustering technique has five steps, including:

- (1) Determine the number of clusters;
- (2) Initialize cluster centroids (average) or “mean” randomly. Centroid is the center point of the cluster;
- (3) Calculate the distance between the data and the centroid using the Euclidean Distance equation:

$$d(P, Q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

- (4) Group the data in the cluster with the closest or minimum distance from each data to the centroid; and
- (5) Calculate the new centroid value using the formula:

$$\text{New centroid} = \frac{\text{Summing all values in each cluster}}{\text{The amount of data in that cluster}}$$

Table 1. Occurrence of words in respect of each theme

Theme	Occurrence of the Word	Relation
Marine	23,275	4
Plastics	17,467	3
Effect	13,043	3
Water	12,816	3
Waste	12,235	2
Sediments	11,711	2
Microplastics	11,468	3
Area	10,615	3
Surface	3,563	2
Sites	2,775	3
Plants	2,426	3
Treatment	1,865	3
Global	1,844	2
Soil	1,778	3
Polyphenylene Ether	1,013	2
Ecosystem	947	1

Step 1 must be done to normalize the data from the occurrence of words and the relationship between these words and others. Table 1 describes the words used by the articles and how many words are related to other words. For example, for the word “marine”, the appearance of the word is around 23,275 and it is related to other words as

much as 4.

Before describing or performing a normalization process on Occurrence of the Word and Relation, the present study first determined the minimum and maximum numbers of occurrence of a theme word and its relationship with other words. Here are the minimum and maximum Occurrences of the Words and their Relation as reflected in Table 1.

Minimum Occurrence of the Word: 947

Maximum Occurrence of the Word: 23,275

Minimum Relation: 1

Maximum Relation: 4

The normalization process was used to give the same proportion, so that it could facilitate the clustering process to ensure the clustering results were not affected by the difference in scale between features. The Normalization formula (Saputra & Kristiyanti, 2022) used is:

$$Age_{new} = \frac{Age_{old} - Age_{min}}{Age_{max} - Age_{min}}$$

Table 2 describes a calculation process to normalize the Occurrence of the Word. For example, the theme “marine” had 23,275 Occurrence of the Word and if entered in a formula, it will become:

$$Marine = \frac{23275 - 947}{23275 - 947} = 1$$

Table 2. Normalization of the occurrence of a theme word

Theme	Occurrence of the Word
Marine	1
Plastic	0.74
Effect	0.54
Water	0.53
Waste	0.51
Sediments	0.48
Microplastics	0.47
Area	0.43
Surface	0.12
Sites	0.08
Plants	0.07
Treatment	0.04
Global	0.04
Soil	0.04
Polyphenylene Ether	0.003
Ecosystem	0

Table 3. Normalization of relation

Theme	Relation
Marine	1
Plastics	0.7
Effect	0.7
Water	0.7
Waste	0.3
Sediments	0.3
Microplastics	0.7
Area	0.7
Surface	0.3
Sites	0.7
Plants	0.7
Treatment	0.7
Global	0.3
Soil	0.7
Polyphenylene Ether	0.3
Ecosystem	0

Table 3 describes a calculation process for normalizing relations. For example, the theme “marine” had four numbers of relation and if entered in a formula, it will become:

$$Marine = \frac{4-1}{4-1} = 1$$

The next step is to determine the centroid on the x- and y-axes in the three clusters, the new cluster, and the value of the Sum of Squared Error (SSE).

(1) Iteration 1

Iteration in the *k*-means algorithm refers to the steps taken to generate the final cluster. The initial step in performing an iteration starts with determining the centroid on the x- and y-axes. To determine the centroid on the x-axis and y-axis in manual calculations with Microsoft Excel, the formula for determining the centroid on the x-axis is: average if (number of initial clusters; centroid on the x-axis; number of relation). The formula for determining the centroid on the y-axis is: average if (number of initial clusters; centroid on the y-axis; number of occurrences of the word); the results of calculations using these formulas can be presented in Table 4.

Table 5 describes the assessment to determine the new cluster. In determining the new cluster, there are several manual calculations using Microsoft Excel, including: The distance between each data point and the centroid was calculated using the squared Euclidean distance:

$$d_{ij} = (x_i - c_{jx})^2 + (y_i - c_{jy})^2$$

where, x_i and y_i represent the normalized features, and c_{jx} , c_{jy} represent the centroid coordinates of cluster j . Each data point is assigned to the cluster with the minimum distance, defined as:

$$Cluster(x_i) = arg_j \min d_{ij}$$

After getting a new cluster, the next step is to calculate the SSE value using the SSE equation. SSE is a metric used to measure the extent to which data points in a cluster are scattered or distant from the cluster center. The manual calculation in determining the SSE value is the sum of the squared distance values.

Table 4. Determining the initial centroid

Centroid	1	2	3
x	0.5	0.5	0.6
y	0.3	0.5	0.2

Table 5. Determining the new cluster

Squared Distance	1	2	3	New Cluster
0.6	0.7	0.6	0.8	2
0.1	0.2	0.1	0.3	2
0.0	0.1	0.0	0.1	2
0.0	0.1	0.0	0.1	2
0.0	0.1	0.0	0.2	2
0.0	0.1	0.0	0.1	2
0.0	0.0	0.0	0.1	2
0.0	0.0	0.0	0.0	1
0.1	0.1	0.1	0.1	1
0.0	0.1	0.2	0.0	3
0.0	0.1	0.2	0.0	3
0.0	0.1	0.2	0.0	3
0.1	0.1	0.2	0.1	1
0.0	0.1	0.2	0.0	3
0.1	0.1	0.2	0.1	1
0.4	0.4	0.4	0.4	1

(2) Iteration 2

Table 6 describes a manual calculation in determining the centroid in iteration 2, following the same steps of calculation as in iteration 1. Table 7 describes a calculation to determine the new cluster in iteration 2 with the same calculation for the new cluster as in iteration 1.

Table 6. Determining the centroid in iteration 2

Centroid	1	2	3
x	0.2	0.6	0.6
y	0.1	0.6	0.0

Table 7. Determining the new cluster in iteration 2

Relation (X)	Occurrence of the Word (Y)	Cluster
1.0	1.00	2
0.7	0.74	2
0.7	0.54	2
0.7	0.53	2
0.3	0.51	2
0.3	0.48	2
0.7	0.47	2
0.7	0.43	2
0.3	0.12	1
0.7	0.08	3
0.7	0.07	3
0.7	0.04	3
0.3	0.04	3
0.7	0.04	3
0.3	0.00	3
0.0	0.00	1

(3) Iteration 3

Table 8 describes a manual calculation in determining the centroid in iteration 3, following the same steps as in iteration 1. Table 9 describes a calculation to determine the new cluster in iteration 3 with the same calculation as in iteration 1.

Table 8. Determining the centroid in iteration 3

Centroid	1	2	3
<i>x</i>	0.25	0.6	0.7
<i>y</i>	0.040073	0.6	0.1

Table 9. Determining the new cluster in iteration 3

Squared Distance	1	2	3	Cluster
0.31	1.48	0.31	1.00	2
0.02	0.66	0.02	0.47	2
0.00	0.43	0.00	0.24	2
0.00	0.42	0.00	0.23	2
0.09	0.22	0.09	0.31	2
0.10	0.20	0.10	0.29	2
0.02	0.36	0.02	0.17	2
0.03	0.33	0.03	0.14	2
0.01	0.01	0.31	0.11	1
0.00	0.18	0.26	0.00	3
0.00	0.17	0.27	0.00	3
0.00	0.17	0.30	0.00	3
0.01	0.01	0.39	0.11	1
0.00	0.17	0.31	0.00	3
0.01	0.01	0.43	0.11	1
0.06	0.06	0.74	0.45	1

3.4 Structural Graph Metrics

To strengthen the structural claims of the semantic network, multiple graph metrics were computed. The node “marine” held the highest degree centrality (4) and closeness centrality, thus confirming its central role within the conceptual system. High betweenness values were observed for “ecosystem”, “treatment”, and “PPE”, indicating their roles as semantic intermediaries connecting thematic domains. Cluster-level cohesion scores confirm microplastics as the densest cluster, followed by plastics and soil. These metrics quantitatively support the thematic segmentation derived from the ACA and DAG visualization.

4. Discussion

The combination of ACA and graph theory within the SLR framework offers an innovative approach for examining environmental datasets on a large scale. The findings indicate a shift in the study of marine debris from

merely describing pollution types to employing more sophisticated computational and network-based modelling techniques. The use of Leximancer allowed the extraction of semantically coherent concepts, and the graph-based analysis helped identify relational structures among these concepts. Three prominent thematic clusters were identified, microplastics, plastics, and soil, with each illustrating distinct environmental pathways of marine debris pollution. The microplastics cluster underscores the enduring presence and bioaccumulation of tiny plastic particles in aquatic ecosystems, thus drawing attention to their ecological and toxicological hazards. The plastics cluster embodies international discussions surrounding macro-debris management, waste policies, and their effects on ecosystems, in accordance with contemporary sustainability frameworks established by the United Nations Decade of Ocean Science. The soil cluster, while not the most prominent, highlights the increasing focus on land-based contributors to marine pollution and the interconnectedness of terrestrial and marine ecosystems.

Analyzing graphs through DAG and adjacency matrices offered valuable structural insights into the interconnections of research concepts. The primary nodes “marine”, “plastics”, and “waste” demonstrated the greatest degree and closeness centrality, serving as key semantic hubs within the body of research. Conversely, emerging nodes like “treatment”, “ecosystem”, and “PPE” exhibited lower centrality yet higher betweenness, indicating their function as bridging concepts that linked various thematic domains. The application of normalization and *k*-means clustering clarified the relational intensity among terms, demonstrating that conceptual proximity corresponded with thematic coherence. The findings confirm earlier bibliometric observations that research on marine debris was evolving towards multi-scalar and interdisciplinary approaches that incorporated environmental monitoring, data analytics, and policy frameworks. Furthermore, the integration of ACA and graph theory enhanced reproducibility and objectivity, thereby minimizing the subjectivity often found in conventional literature reviews. This study demonstrated that computational methods, particularly when enhanced by graph-based modelling, provided significant benefits in synthesizing intricate research ecosystems. They enabled the visualization of concealed conceptual networks, assisted in the generation of automated hypothesis, and created new avenues for research in environmental informatics centered on sustainability and marine conservation.

4.1 Strengthened Scientific Contributions

This study contributes to environmental informatics by demonstrating how ACA and graph theory could be integrated to reveal hidden semantic structures in large-scale environmental literature. Beyond descriptive mapping, the incorporation of the DAG validation and structural graph metrics enabled the detection of hierarchical and bridging concepts. This contributes a methodological advancement over traditional SLRs that rely solely on narrative synthesis or frequency counts. The resulting framework offers a replicable approach for detecting emerging research fronts and cross-domain interactions in sustainability science.

5. Conclusions

This study introduced a cohesive analytical framework that merged ACA with graph theory to perform a SLR which focused on marine debris. Through the examination of 357 studies indexed in Scopus, three primary thematic clusters emerged, i.e., microplastics, plastics, and soil, with each illustrating unique yet interrelated aspects of marine pollution. The use of graph-based modelling indicates that the conceptual framework of marine debris literature was primarily influenced by high-centrality nodes like “marine”, “plastics”, and “waste”, which formed the core of the global research network. The findings indicate that the integration of ACA and graph theory improved the objectivity, accuracy, and scalability of synthesizing environmental knowledge. This hybrid method enabled the detection of structural and semantic relationships that were frequently overlooked in manual reviews, thus offering a data-driven foundation for pinpointing research gaps and developing sustainable solutions. This approach fosters methodological advancement in environmental informatics, via providing a model that could be replicated in other areas like climate change, biodiversity, and waste management. Future investigations should broaden this framework by integrating dynamic network analysis, topic modelling, and machine learning to track the temporal evolution of environmental knowledge systems. The integration of semantic automation with network analytics signifies a compelling avenue for progressing in the field of computational sustainability.

The study advanced both methodological and substantive understanding of marine debris research. Methodologically, it provided a reproducible hybrid ACA graph framework for mapping and quantifying conceptual structures. Substantively, it identified three robust thematic domains and highlighted underexplored bridging concepts that represented potential research gaps. This dual contribution positions the study as a foundation for computational approaches to the synthesis of environmental knowledge.

5.1 Limitations

Despite a structured and automated mapping of marine debris research, this study is not without limitations. First, the exclusive reliance on Scopus-indexed articles may introduce database bias because Scopus prioritizes

journals with particular indexing criteria, potentially underrepresenting regional studies, non-English literature, and emerging interdisciplinary contributions. As a result, certain context-specific insights or innovative early-stage works may not be captured in the dataset. Second, the Leximancer engine used for ACA primarily identifies high-frequency concepts and this means that low-frequency but strategically important emerging terms like “biodegradable plastics”, “nanoplastics”, or “AI-driven monitoring” may be overlooked. This limitation is inherent to frequency-driven semantic extraction and could marginalize weak but growing research signals within the corpus.

5.2 Directions of Future Research

Future research should address these limitations by expanding the corpus to include additional databases such as Web of Science, Dimensions, or Google Scholar, which may provide a broader disciplinary spectrum and reduce database-specific bias. Integrating multiple bibliographic sources would improve representativeness and allow comparative analyses across indexing systems. Methodologically, subsequent studies may incorporate complementary computational techniques such as Latent Dirichlet Allocation (LDA), BERTopic, or Bidirectional Encoder Representations from Transformers (BERT)-based semantic clustering to identify low-frequency but high-relevance emerging concepts that Leximancer may omit. Moreover, hybrid frameworks that combine ACA with temporal network analysis, evolution of dynamic topics, or detection of machine learning-based concept could offer deeper insights into how environmental themes evolve over time. Such methodological enhancements would strengthen the robustness, scalability, and predictive value of research on marine debris and related environmental challenges.

Author Contributions

Research methodology, M.A.M. and S.; graph theory contribution, Z.A. and F.S.F.K.; writing—original draft preparation, H.; writing—review and editing, M.A.M., S., Z.A., and F.S.F.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

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Conflicts of Interest

The authors declare no conflict of interest.

References

- Agamuthu, P., Mehran, S. B., Norkhairah, A., & Norkhairiyah, A. (2019). Marine debris: A review of impacts and global initiatives. *Waste Manag. Res.*, 37(10), 987–1002. <https://doi.org/10.1177/0734242X19845041>.
- Al-Khalidi, M., Al-Zaidi, R., Woods, J., Reed, M., & Pereira, E. (2019). Securing marine data networks in an IoT environment. In *2019 7th International Conference on Future Internet of Things and Cloud (FiCloud), Istanbul, Turkey* (pp. 125–132). <https://doi.org/10.1109/FiCloud.2019.00025>.
- Bertolazzi, S., Cuttitta, A., & Pipitone, V. (2024). Addressing marine plastic pollution : A systematic literature review. *Curr. Opin. Environ. Sustain.*, 68, 101428. <https://doi.org/10.1016/j.cosust.2024.101428>.
- Boote, D. N. & Beile, P. (2005). Scholars before researchers: On the centrality of the dissertation literature review in research preparation. *Educ. Res.*, 34(6), 3–15. <https://doi.org/10.3102/0013189X034006003>.
- Chang, Y. C. & Saqib, M. (2025). Impacts of the global plastic treaty on the marine environmental protection law of China. *Water*, 17(11), 1633. <https://doi.org/10.3390/w17111633>.
- Chao, H. C., Wu, H. Te, & Tseng, F. H. (2021). AIS meets IoT: A network security mechanism of sustainable marine resource based on edge computing. *Sustainability*, 13(6), 3048. <https://doi.org/10.3390/su13063048>.
- Daniel, F. & Taneo, P. N. L. (2019). *Teori Graf*. Yogyakarta: Deepublish.
- Derenevich, M. G., Yasue, J. E., Bitencourt, R. S., & de Macedo Guimarães, L. B. (2022). Macroergonomics: An analysis of the definition based on the literature. *Rev. Ação Ergon.*, 16(2), e202205.

- <https://doi.org/10.4322/rae.v16n2.e202205>.
- Eto, M. (2019). Extended co-citation search: Graph-based document retrieval on a co-citation network containing citation context information. *Inf. Process. Manage.*, 56(6), 102046. <https://doi.org/10.1016/j.ipm.2019.05.007>.
- Evans, M., McIntosh, W., Lin, J., & Cates, C. (2007). Recounting the courts? Applying automated content analysis to enhance empirical legal research. *J. Empir. Leg. Stud.*, 4(4), 1007–1039. <https://doi.org/10.1111/j.1740-1461.2007.00113.x>.
- Fulke, A. B., Bhanushali, S., & Jadhav, H. S. (2026). Global marine plastic pollution: Sources, distribution, implications on human health and mitigation strategies. *Cont. Shelf Res.*, 296, 105578. <https://doi.org/10.1016/j.csr.2025.105578>.
- Kosior, E. & Crescenzi, I. (2020). Solutions to the plastic waste problem on land and in the oceans. In *Plastic Waste and Recycling: Environmental Impact, Societal Issues, Prevention, and Solutions*. Elsevier. <https://doi.org/10.1016/B978-0-12-817880-5.00016-5>.
- Marlina, A., Fazriansyah, D. T., Bimo, W. A., Sinaga, H. Z., Maulana, H., & Ritzkal (2024). A comprehensive guide to bibliometric analysis for advancing research in digital business. *J. Intell. Manag. Decis.*, 3(3), 187–201. <https://doi.org/10.56578/jimd030305>.
- Pachayappan, M. & Venkatesakumar, R. (2018). A graph theory based systematic literature network analysis. *Theor. Econ. Lett.*, 8(5), 960–980. <https://doi.org/10.4236/tel.2018.85067>.
- Purba, N. P., Handyman, D. I. W., Pribadi, T. D., Syakti, A. D., Pranowo, W. S., Harvey, A., & Ihsan, Y. N. (2019). Marine debris in Indonesia: A review of research and status. *Mar. Pollut. Bull.*, 146, 134–144. <https://doi.org/10.1016/j.marpolbul.2019.05.057>.
- Puskic, P. S., Cramer, I., Church, E., Deery, E., Egger, M., Fox, N., De Haan, W. P., Lebreton, L., Liconti, A., Sanchez-Vidal, A., et al. (2026). Highly engaged marine users can help monitor marine plastic pollution in under accessed environments. *Ocean Coast. Manag.*, 272, 107983. <https://doi.org/10.1016/j.ocecoaman.2025.107983>.
- Puspitasari, D., Syukur, A., Fanani, A. Z., & Hasibuan, Z. A. (2024). Theoretical framework on automatic content analysis approach to arrhythmia clustering. *Int. Rev. Model. Simul.*, 17(5), 348–357. <https://doi.org/10.15866/iremos.v17i5.25236>.
- Rahmah, A., Santoso, H. B., & Hasibuan, Z. A. (2022). Critical review of technology-enhanced learning using automatic content analysis. *Int. J. Adv. Comput. Sci. Appl.*, 13(1), 385–394. <https://doi.org/10.14569/IJACSA.2022.0130148>.
- Ritzkal, Muljono, Santoso, H. A., & Hasibuan, Z. A. (2023). Theoretical framework review of plastic waste management. *J. Namib. Stud.*, 33, 1774.
- Saputra, I. & Kristiyanti, D. A. (2022). *Machine Learning Untuk Pemula* (1st ed.). Informatika.
- Sauer, P. C. & Seuring, S. (2023). How to conduct systematic literature reviews in management research : A guide in 6 steps and 14. *Rev. Manag. Sci.*, 17(5), 1899–1933. <https://doi.org/10.1007/s11846-023-00668-3>.
- Schrodt, J., Dudchenko, A., Knaup-Gregori, P., & Ganzinger, M. (2020). Graph-representation of patient data: A systematic literature review. *J. Med. Syst.*, 44(4). <https://doi.org/10.1007/s10916-020-1538-4>.
- Sutriawan, S., Rustad, S., Shidik, G. F., Pujiono, P., & Muljono, M. (2024). Review of ambiguity problem in text summarization using hybrid ACA and SLR. *Intell. Syst. Appl.*, 22, 200360. <https://doi.org/10.1016/j.iswa.2024.200360>.
- Webster, J. & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *Manage. Inf. Syst. Q.*, 26(2), 13–23. <https://doi.org/10.2307/4132319>.
- Widiastutie, S., Maarif, D., Saraswati, D. P., Sianipar, I. M. J., Phan, T. T. T., & Nguyen, V. V. (2025). Residents' adaptive marine plastic litter management in Galang Island, Indonesia: An importance-performance analysis. *Mar. Policy*, 172, 106508. <https://doi.org/10.1016/j.marpol.2024.106508>.